INCH-POUND

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MILITARY SPECIFICATION

TRANSMITTER, RATE OF FLOW TRU-6/A, 05,000-PPH

The specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of 0-5,000 pph rate of flow transmitter, designated TRU-6/A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-1672	Boxes, Shipping, Reusable With Cushioning

MILITARY

MIL-P-116	Preservation, Methods Of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/ TICLA, Tinker AFB, OK 73145-5990 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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MIL-B-121	Barrier Material, Greaseproofed. Waterproofed Flexible
MIL-C-5541	Chemical Films and Chemical Film Materials For Aluminum And Aluminum Alloys
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-A-8625	Anodic Coatings For Aluminum And Aluminum Alloys
MIL-I-26299	Indicator, Rate Of Flow EFU-2/A, 0-5, 000 PPH, Integrally Lighted
MIL-B-27497	Bearings, Jewel, Sapphire Or Ruby, Synthetic
MIL-M-45202	Magnesium Alloys, Anodic Treatment Of

NAVAL AIR SYSTEMS COMMAND

SD-24	General Specifications For Design And Construction Of Aircraft Weapon Systems

STANDARDS

FEDERAL

FED-STD-101	Test Procedures For Packaging Materials

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipments
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-704	Electric Power, Aircraft Characteristics And Utilization Of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-810C	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MIL-STD-970	Standards & Specifications, Order Of Preference For The Selection Of
MIL-STD-2073-1	DOD Material Procedures For Development And Application Of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specification, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Building #4, Section D, Philadelphia, PA 19111-5094.)

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specification, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.3), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.3 Materials.

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the transmitter except where magnetic materials are essential.

3.3.2 Nonferrous materials. Nonferrous materials shall be used for all parts of the transmitter except where ferrous materials are essential.

3.3.3 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

3.3.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.5 Toxic and corrosive fumes. The transmitter shall be of such material and construction that when the material is burned, fumes or gases that combine with the atmosphere to form acid or corrosive alkali will not be liberated.

3.3.6 Protective treatment. When materials are used in the construction of the transmitter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.7 Titanium. Where high temperature and other stringent environmental conditions are a factor in the design of the transmitter, the use of titanium will be permitted, with the approval of the procuring activity where lightweight and highstrength materials are required. Use of lightweight materials and weight-saving designs shall be of major consideration and their use shall be investigated and exploited to the greatest possible extent. For Navy use, titanium (magnesium) shall conform to SD-24.

3.3.8 Cadmium is not to be used in the transmitter unless approved by the preparing activity. There shall be no cadmium used on the exterior of the transmitter. The fluid whose flow rate is being measured by the transmitter shall not come into contact with any cadmium.

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3.4 Design and construction. The transmitter shall be calibrated for rate of flow in pounds per hour of aviation fuel conforming to MIL-T-5624. The specific gravity of the fuel shall be 0.785 ± 0.005 at 15.6°C. A commercial fuel of higher flash point may be used.

3.4.1 Compatibility and interchangeability. The TRU-6/A transmitters shall be interchangeable with previously manufactured TRU-6/a transmitters and shall be completely compatible with the EFU-2/A indicator conforming to MIL-I-26299.

3.4.2 Transmitter. The transmitter shall be designed to operate within the tolerances specified herein with an EFU-2/A indicator conforming to MIL-I-26299. The transmitter shall provide a synchro output to operate Clifton Precision Products Company's type CRC-8-A-9 synchro torque receiver, or equal. For an increasing flow rate, the direction of rotation of the transmitter synchro shaft shall be counterclockwise as viewed from the shaft end.

3.4.2.1 Indicator. When two EFU 2/A indicators are operated from one transmitter, the resulting errors of the indicator synchros shall not exceed 1° of are over the errors resulting when only one indicator is used.

3.4.3 Power. The transmitter shall be designed to operate within the tolerances specified herein from 115V±10 percent, 400 Hz±25 percent, single-phase power conforming in other characteristics to MIL-STD-704, except that the transmitter motor which receives regulated power from the EFU-2/A indicator shall be designed to operate from 65 to 126V 400±1 Hz power. The transmitter synchro shall operate with voltage surges to 195±5V as specified in 4.6.10.

3.4.4 Range. The transmitter shall be designed to measure flow rates from 0 to 5,000 pph.

3.4.5 Mechanical stop. A mechanical stop shall be provided within the transmitter above the 5,000-pph point but below the zero point.

3.4.6. Pressure drop. The transmitter shall be so designed that when inoperative, the pressure drop will not be appreciably more under any fuel flow conditions than when it is operating normally.

3.4.7 Reliability and longevity.

3.4.7.1 Reliability program. The contractor shall establish a reliability program in accordance with MIL-STD-781 (see 6.2.2).

3.4.7.2 Mean-time-between-failures (MTBF). The transmitter shall have a minimum acceptable MTBF of 1,000 hours demonstrated to a 90 percent confidence level.

3.4.7.3 Longevity. The transmitter shall have an operating life span of not less than 2,500 hours before wearout failures occur or the transmitter consistently fails to meet the specified MTBF index. parts requiring replacement within this period of time and the normal operating period of such parts shall be reported by the contractor to the procuring activity.

3.4.8 Maintainability. The contractor shall establish a maintainability program in accordance with MIL-STD-470 which shall include specific design objectives for mean corrective maintenance downtime (M_{ct}), and maximum corrective maintenance downtime ($M_{max_{ct}}$), confidence and percentile levels, and equipment repair time. The following maintainability requirements shall be considered in the design of the transmitter:

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a. Minimization of the complexity of maintenance tasks (for example: calibration, adjustments, inspection, etc.) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items.

b. Rapid and positive recognition of equipment malfunction or marginal performance.

c. Rapid and positive identification of the replaceable defective part, assembly, or component.

d. Minimum numbers and types of tests and test equipment (special and standard) needed to perform maintenance.

e. Optimum accessibility of all components requiring maintenance inspection, removal, or replacement.

3.5 Performance. The transmitter shall be capable of meeting the requirements specified herein under the following conditions:

a. Temperatures: Operation at temperatures ranging from -54° to +190°C and exposure to temperatures ranging from -62° to +190°C with fuel ambient temperatures through the transmitter ranging from -54° to +71°C.

b. Vibration: Vibration from 0.036 double amplitude to ±20g at frequencies from 5 to 2,000 Hz.

c. Altitude: Altitudes ranging from sea level to 100,000 feet.

d. Shock: Shock forces up to 20g for 11 milliseconds.

e. Salt fog: Salt fog for a period of 50 hours.

f. Humidity: Relative humidity up to 95 percent.

g. Fungus: Fungus growth as encountered in tropical climates.

h. Voltage and frequency variation: Voltage frequency variations ranging from 104V 300 Hz to 126V 500 Hz.

i. Dielectric strength: Application of 500V between any pin on the electrical receptacle and the transmitter case for five seconds.

j. Voltage surge: Voltage surge up to 195V.

k. Frequency cycling: Frequency cycling from 320 to 480 Hz.

l. Response: Response in less than four seconds from zero to full scale.

3.5.1 Operating pressure. The transmitter shall be capable of operating in fuel systems with normal fuel pressure up to 1,000 psi, shall withstand, without damage, a static fuel pressure of 1,600 psi at both room temperature and 54°C, and shall withstand a static fuel pressure of 1,000 psi at a fuel temperature of 71°C when in an ambient temperature of 140°C.

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3.5.2 Electromagnetic interference. The transmitter shall meet the radiated, conducted, and susceptibility electromagnetic interference requirements of MIL-STD-461 for class ID equipment except the frequency range shall be from 0.15 to 150 mc.

3.5.3 Tolerances. The transmitter shall operate within the tolerances specified in tables I, II, and III.

3.5.4 Power consumption. Power consumption of the transmitter shall not exceed 20 va with 115 ± 1V 400 ± 2 Hz excitation when the transmitter is operating and providing an indication of 2,000 pph at room temperature conditions.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

TABLE I. Fuel Flow Transmitter Scale Error Tolerances

	Scale Error Tolerance PPH ±	
Fuel Flow PPH	Room Temperature and Substitute Fuel	High and Low Temperature
200	20	30
500	20	30
1,000	15	25
1,500	15	35
2,000	20	40
2,500	25	40
3,000	30	40
3,500	35	45
4,000	40	50
4,500	45	55
5,000	50	60

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TABLE II. Tests and Tolerances

Test	Tolerance PPH ±
Voltage and frequency variation system basis	1-1/2 percent of point above 500 pph
Position error	15
Friction	
Room temperature	10
High temperature	10
Low temperature	15
Vibration	
Pointer oscillation (double amplitude)	20
Pointer variation	20
Pointer oscillation (double amplitude)	80
Pointer variation	80

TABLE III. Pressure Drop

Flow PPH	Maximum Pressure Drop PSI
2,000	0.6
4,000	2.2
Test to be conducted with end housing adapters supplied with basic transmitter	

3.7 Mounting. The basic transmitter mounting bosses and spacing thereof shall be considered as part of the basic transmitter and shall be shipped with, but not assembled to, each transmitter. Protective leakproof fluid plates or caps shall be attached to the basic transmitter for shipment purposes.

3.8 Flow direction. The direction of fuel flow through the transmitter shall be permanently and legibly marked by an arrow, as shown on figure 1, on both the front and the back of the item.

3.9 Electrical receptacle. The transmitter shall be provided with a hermetically sealed electrical receptacle, Cannon part No. MC14H-10-6P, or equivalent. System connections shall be as shown on figure 3. Where practicable, the internal wiring shall be color coded and the pins shall be wired for the specified connections as follows:

Pin	Color	
A	Black/White	(R2)C2 lead of transmitter synchro rotor, low potential input of 115V 400-Hz regulated power to motor and external ground
B	Red/White	(R1)H2 lead of transmitter synchro rotor (115V 400-Hz input)
C	Blue	(S1) Y2 of transmitter synchro stator
D	Yellow	(S3)X2 of transmitter synchro stator
E	Black	(S2)Z2 of transmitter synchro stator (not connected to C2)
F	Black	115V 400-Hz input from regulated power supply to motor lead.

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3.10 Electrical zero. For the purpose of this specification, the transmitter synchro rotor leads shall be designated C2 and H2. The stator leads shall be labeled X2, Y2, and Z2. The subscripts indicate the requirement for a low null at electrical zero. Identification of the leads and the determination of electrical zero shall be as follows: Arbitrarily label one rotor lead as C2, the low potential or ground lead, and label the other rotor lead H2, the high potential lead. Arbitrarily label one stator lead Z2 and connect Z2 to C2. Apply the rated voltage of the rotor between C2 and H2. Rotate the rotor to obtain the following conditions which indicates electrical zero, and label the other stator leads accordingly:

a. The voltage between the unlabeled stator leads will be a minimum, or at null.

b. The voltage between one of the unlabeled stator leads and the Z2 lead shall increase before decreasing when the rotor is rotated in a clockwise direction from electrical zero as viewed from the shaft end. The maximum shall occur at approximately 30 degrees of rotation. Label this lead X2. Label the remaining stator lead Y2. At electrical zero, the voltages X2 to Z2 and Y2 to Z2 shall be approximately in phase with excitation voltage C2 to H2. Schematically, the stator leads will be designated X2, Y2, Z2 in a counterclockwise direction.

3.11 Synthetic jewel bearings. When synthetic sapphire or ruby jewel bearings are used, they shall conform to MIL-B-27497.

3.12 Dimensions. The outline dimensions shall conform to figure 1.

3.13 Weight. The weight of the complete transmitter shall not exceed five pounds.

3.14 Contaminated fuel. The transmitter shall operate as specified herein with contaminated fuel containing foreign particles of the sizes shown in table IV.

TABLE IV. Contaminated fuel particle size

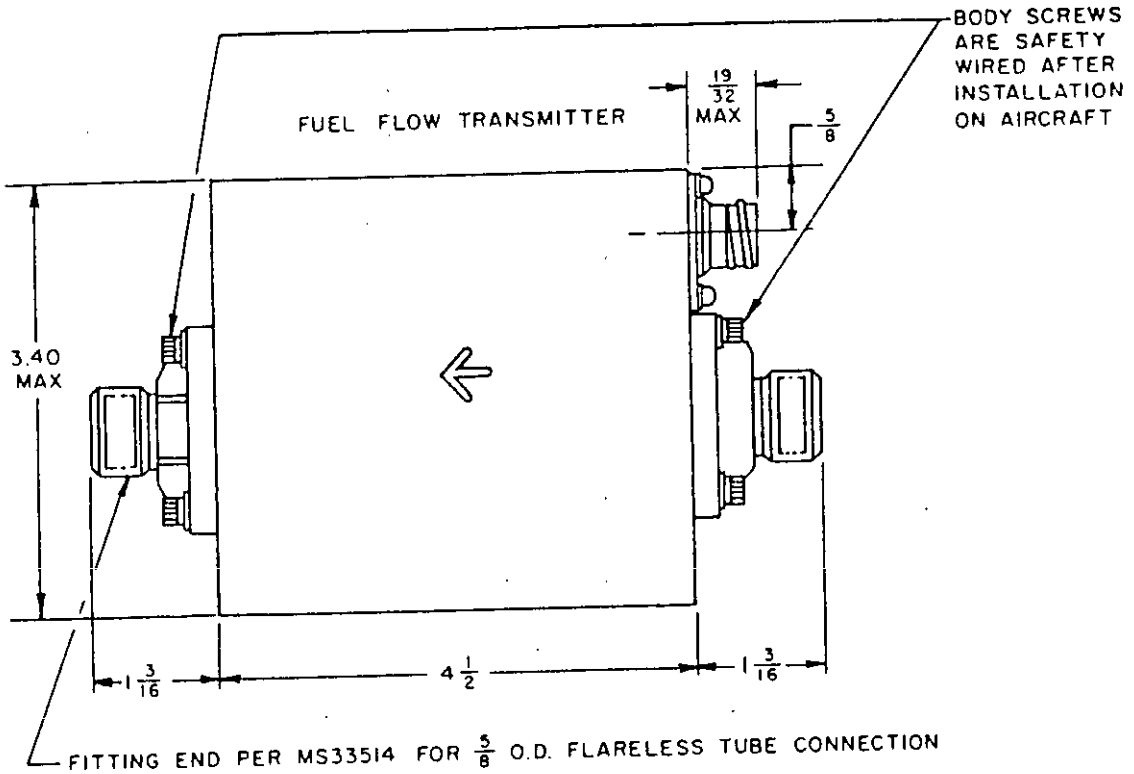
Foreign Particle Size (Microns)	Percent of Total Weight
0-5	39 ±2
5-10	18 ±3
10-20	16 ±3
20-40	18 ±3
over 40	9 ±3
Through a 200 mesh screen	100 percent

3.15 Caps. The fuel-line connections shall be provided with protective shipping caps or plugs to prevent foreign material from entering the transmitter.

3.16 Finishes and protective coatings.

3.16.1 Alloy parts. Where practicable, aluminum or titanium alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Small holes, inserts, and small areas that must be machined after finishing need not be covered with a protective film.

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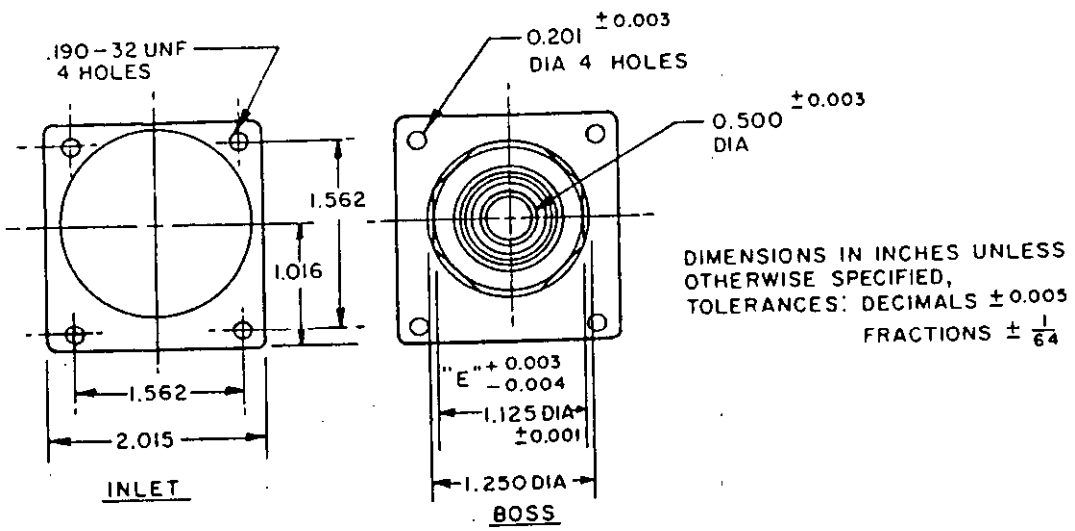
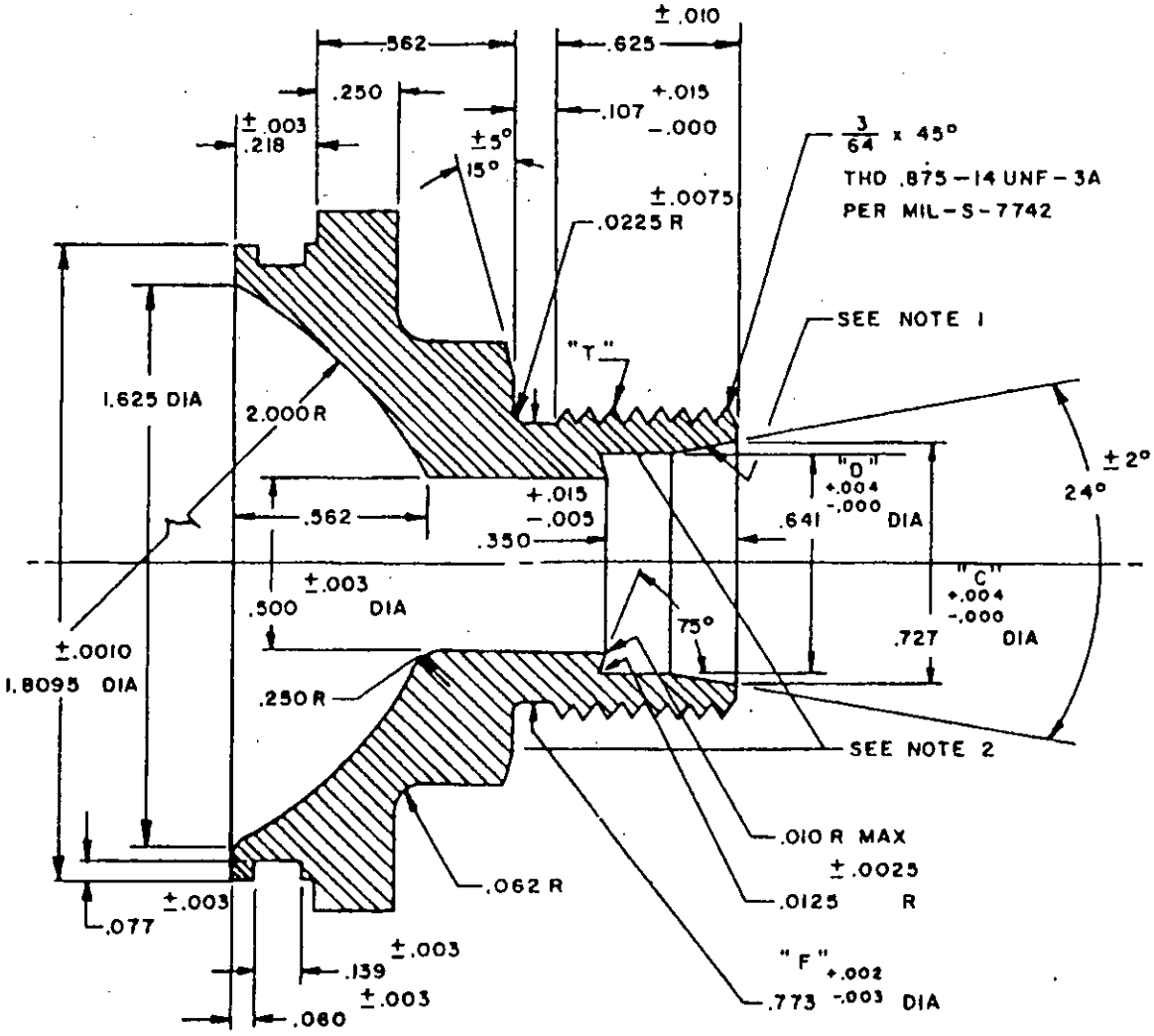


FIGURE 1. Basic transmitter and mounting

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NOTES:

1. This surface shall be free from burrs, longitudinal and spiral tool marks, and shall be smooth, except annular tool marks will be allowed to 63 microinches rms maximum. Styles G and E.

2. This surface shall be free from burrs, longitudinal and spiral tool marks, and shall be smooth, except that annular tool marks will be allowed to 100 microinches rms maximum.

3. Concentricity: D, C, and T (pitch diameter) shall be concentric to each other within 0.008 full indicator reading. F and T (pitch diameter) shall be concentric to each other within 0.005 full indicator reading.

Dimensions in inches. Unless otherwise specified, tolerance: Decimals ±0.016

FIGURE 2. Boss Detail

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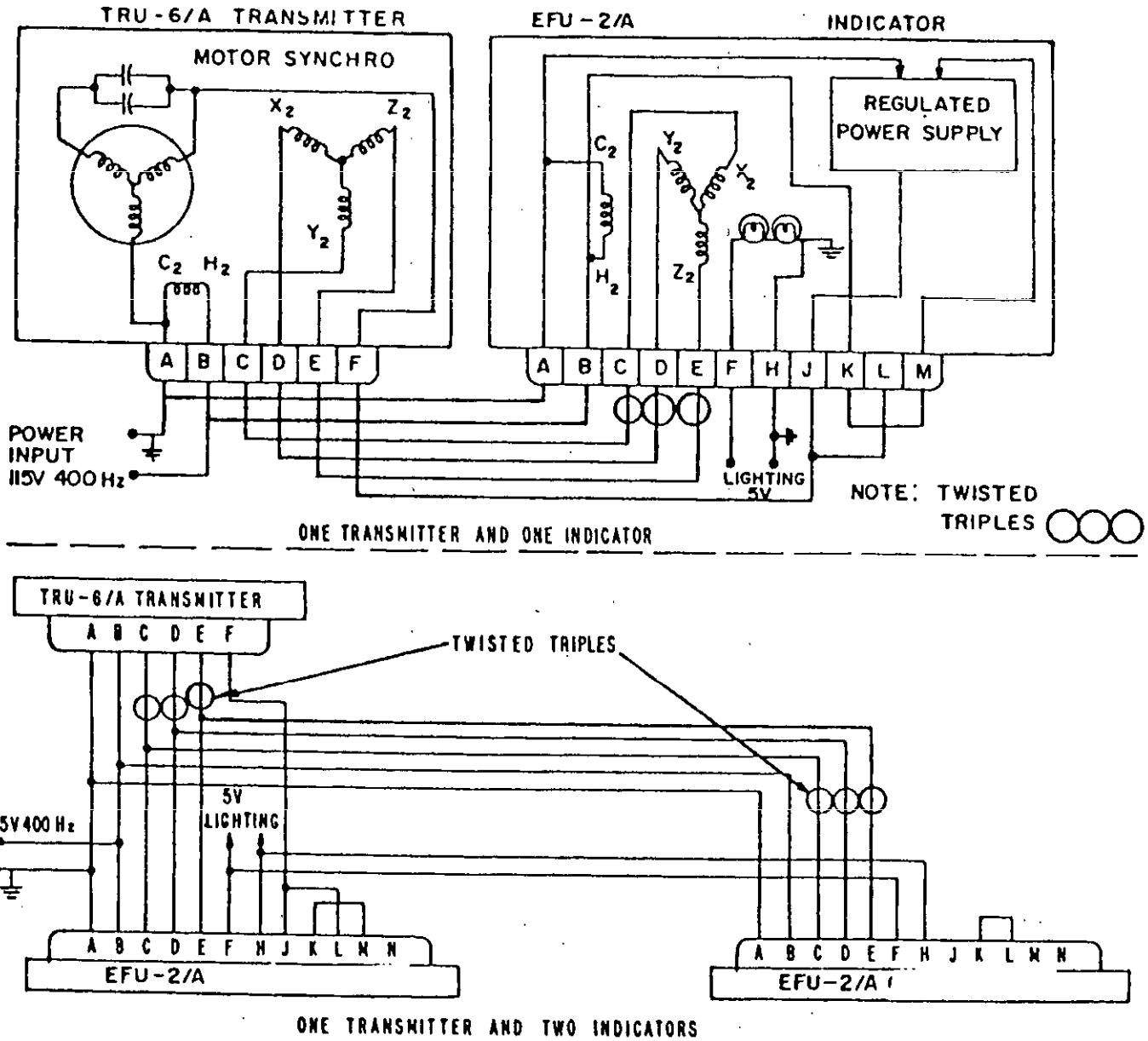


FIGURE 3. System connections

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3.16.2 Chemical treatment. Where the primary purpose of the treatment is to afford a suitable pain base, or for castings containing nonalluminum or titanium alloy inserts, a chemical treatment in accordance with MIL-C-5541 may be used in lieu of an anodic film for alloys that do not anodize satisfactorily.

3.16.3 Abrasion resistance. Where abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used.

3.16.4 Magnesium parts. Magnesium alloy parts shall be covered with an anodic film in accordance with MIL-M-45202, HAE process.

3.17 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.17.1 Nameplate. A nameplate shall be provided in accordance with MIL-STD-130. Metal decalcomanias shall not be used.

3.17.2 Schematic. An electric schematic showing the function of each connector pin shall be permanently attached to the body of the transmitter.

3.18 Workmanship. The transmitter shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, welding, riveting, brazing, plating, painting, machine screw assemblies, and freedom of parts from burrs and sharp edges. Gears shall be of high quality and shall provide a minimum of backlash necessary for smooth operation. Where necessary, antibacklash techniques shall be used.

3.19 Recycled and reclaimed materials. The use of recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS.

4.1. Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspection are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All item shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorized submission of know defective materiel, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified all inspections shall be performed in accordance with the test conditions specified in 4.3.4 of this specification.

4.3.1 Test power. The supply power shall be single-phase ac, 115V±2 percent, and 400 Hz controlled to less than ±1 Hz and of known value.

4.3.2 Test fuel. Unless otherwise specified, the fuel used for calibration in testing shall be JP-4 conforming to MIL-T-5624. The specific gravity of the fuel shall be 0.785 ±0.005 at 15.6°C. (Note: An approved commercial equivalent with higher flash point may be used.)

4.3.3 Test position. Unless otherwise specified, the transmitter shall be tested with the flow axis horizontal and the connector at the 3 o'clock position when viewed from the inlet end.

4.3.4 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the tests are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowances shall be made for the difference from the specified conditions.

4.3.5 Vibration. Unless otherwise specified, the transmitter shall be vibrated before a test reading is taken. An electric vibrator, set at 3,600 Hz with maximum amplitude of 0.002 inch and attached to the flow pipe may be used.

4.3.6 Test equipment. Unless otherwise specified, the transmitter shall be tested while connected to a master indicator and master power supply. The master power supply shall be such that it will have no frequency error, or if a frequency error should exist (not exceeding 1 Hz at 400 Hz), it shall be subtracted from the test data. All flow tests shall be performed on a primary flow stand which measures liquid mass flow rate directly in terms of weight and time.

4.3.7 Tolerances. Test tolerances shall be as specified in tables I, II, and III.

4.4 First article.

4.4.1 Test samples. The test samples shall consist of at least fourteen transmitters representative of the production equipment. Eleven of the samples shall be subjected to the individual, reliability (see 4.6.29.1) and longevity tests only and the other three samples shall be subjected to all tests except reliability and longevity. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 First article tests. The first article shall be consist of all the tests specified under 4.6. Unless otherwise specified, tests shall be conducted in the order determined by the qualifying activity.

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4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests

4.5.1 Individual tests. Each transmitter shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Dielectric strength

c. Scale error

d. Friction error

e. Leakage

f. Response time

g. Burn-in

4.5.2 Sampling tests. A minimum of three transmitters shall be selected at random from each 100 or fraction thereof produced on the contract and subjected to the following tests as described under 4.6:

a. Individual tests

b. Position error

c. Voltage and frequency variation

d. Voltage surge

e. Power consumption

f. Low temperature

g. High temperature

h. Vibration

i. Pressure drop

4.5.2.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The action taken to preclude recurrence. After correction, the tests specified by the procuring activity shall be repeated.

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4.5.2.1.1 Reconditioning of tested transmitters. Unless otherwise specified (see 6.2), transmitters that have been subjected to the sampling tests shall be reconditioned by the contractor by replacing all worn or damaged parts. After reworking, the contractor shall resubmit the transmitters for acceptance.

4.5.2.2 Individual tests may continue. Individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that all items meet all requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted (see 6.2). If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.6 Test methods.

4.6.1 Examination of product. The transmitter shall be inspected to determine compliance with the requirements specified herein with respect to materials, finish workmanship, and marking. Each unit shall be weighed.

4.6.2 Dielectric strength. A potential of 500V at a commercial frequency shall be applied between any pin and the transmitter case for a minimum of five seconds, there shall be no insulation breakdown as a result of this test.

4.6.3. Scale error. The scale error of each transmitter shall be determined at the test points specified in table I at room temperature. The scale error shall be determined for both increasing and decreasing flow rates. Each test point in a set of readings shall be consistently approached from the same direction. A minimum of two sets of reading shall be taken at each test point. A set of readings shall consist of two flowstand weigh cycle times averaged to obtain average flow rates for two independent runs. The values of both runs shall be reported. If the individual readings for one of the runs exceeds or is within five percent of the specification limit, a second set of readings shall be taken. Scale error shall be determined while the transmitter is gently vibrated by an electromechanical

vibrator mounted on the flow piping. Scale error tests conducted at room, low, and high temperatures, and substitute fuel operating conditions shall be in accordance with this procedure. Test tolerances shall be in accordance with table I.

4.6.4 Friction error. Friction error shall be determined at 25, 50, and 90 percent of full scale. The difference in readings taken before vibration and after vibration shall not exceed the values listed in table II. This test shall be combined with the scale error test.

4.6.5 Leakage. The transmitter shall be tested for leakage as follows: A nonshock hydraulic pressure shall be applied to the inlet connection with the outlet connection bled free of air and capped. A pressure of 1,600 psi shall then be rapidly applied five times a room temperature. After the last application of pressure and while still at the test pressure and temperature, there shall be no indication of leakage for a period of five minutes as checked on a pressure gage readable to ten psi or by visual leakage of a colored test fluid. The transmitter temperature shall then be reduced to -54±2°C by submersion in a cold liquid bath, already at this temperature, for a period of not less than 15 minutes. While still at this temperature, the test pressure shall be rapidly applied five times. After the last application of pressure, there shall be no evidence of leakage for a period of five minutes as detected by the same means used for the room temperature portion of this test.

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4.6.6 Response time. The time required for the transmitter output signal to come within one percent of the final steady state signal shall not exceed 4.0 seconds when a step change in flow rate is applied. The flow rate shall be changed from approximately two percent of full scale to a full scale indication by the activation of a quick opening valve. The timer will be started simultaneously with the activation of the valve.

4.6.7 Burn in. The transmitter shall be placed in a chamber with the temperature maintained at 100°C for a minimum of four hours. The temperature shall then be lowered and maintained at 50°C for 20 hours with fuel flowing equivalent to a half-scale indication. At least the first 8 hours of this test shall be performed during normal working hours. Observations shall be made and recorded at least once every two hours throughout the eight-hour period. At no time shall the transmitter output change by more than one-half the tolerance specified for this test point in the applicable tolerance table. Pointer oscillation or erratic movement shall be cause for rejection. The transmitter shall be returned to room temperature for a minimum of two hours after which it shall meet the test specified in 4.6.3. If the burn-in test is conducted other than as specified herein, the test shall be subject to the approval of the procuring activity.

4.6.8 Position error. Position error shall be determined at 50 percent of full scale. Readings shall be taken in the normal position and with the transmitter turned 90° in any direction. The reading taken in any position shall not differ from the reading taken in the normal position by more than the values specified in table II.

4.6.9 Voltage and frequency variation. At pins B and A, the applied voltage and frequency shall be set at the following points:

	Voltage	Frequency (Hz)
Normal condition	115	400
	104.0	300
	104.0	500
	126.0	300
	126.0	500

This test shall be conducted at 25, 50, and 90 percent of full scale flow rates. Readings shall be taken at the 115V, 400Hz normal condition and at the conditions specified herein. The readings taken at each of the above conditions shall not differ from the readings taken at the normal condition in excess of the value shown in table II. The transmitter shall be operated at least 1/2 hour at each of the specified conditions before any readings are taken. These tests shall be run with a master indicator and a master power supply (see 4.3.6)

4.6.9.1 Pins. The test specified in 4.6.9 shall be repeated with power applied to pins A and F and with the voltage and frequency set as follows:

	Voltage	Frequency (Hz)
Normal condition	115	400
	65	400
	126	400

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4.6.10 Voltage surge. A master indicator need not be used during the surge application of this test. A synchro receiver, or simulated receiver, compatible with the transmitter will be adequate. A master power supply must be used. The test shall be made at approximately 50 percent of full scale. Power shall be applied at 115±1V, 400±2 Hz for 1 hour. At the end of this operating period, the power input shall be adjusted to 115±1V, 320±2Hz. A voltage surge shall then be applied to provide an rms peak voltage of 160 ±5V and return to 115±1V. The duration of the surge including rise and decay time shall be 0.09±.005 second. The power input shall then be adjusted to 115±1V, 480 ±2 Hz and a voltage surge applied to provide an rms peak voltage of 195±5V and return to 115±1V. The duration of this surge including rise and decay time shall be 0.07 ±.005 second. At the conclusion of this procedure, a scale error test at room temperature shall be conducted. The errors shall not exceed the room temperature tolerances of table I.

4.6.11 Power consumption. With 115V 400 Hz and power from the indicator to transmitter motor applied to the transmitter, power consumption of the transmitter shall not exceed 20 va at a flow rate of 2,000 pph.

4.6.12 Low temperature. The transmitter shall be placed in a test chamber and maintained at a temperature of -54°±2°C for a period of four hours. At the end of this period and while still at the specified low temperature, the transmitter shall be subjected to the tests specified in 4.6.3 and 4.6.4. The errors shall not exceed the tolerances specified in tables I and II, respectively.

4.6.13 High temperature. After the chamber and transmitter have been raised to a temperature of 121°±2°C, the transmitter shall be operated at that temperature for six hours. While at the specified high temperature, the transmitter shall be subjected to the tests specified in 4.6.3 and 4.6.4. The errors shall not exceed the tolerances specified in tables I and II, respectively.

4.6.14 Vibration. The vibration test shall be conducted in accordance with MIL-STD-810C, method 514, equipment category (b), curve H, at room temperature with fuel flowing at approximately 50 percent of full scale. Pointer oscillation and variation shall not exceed the tolerance specified in table II for 10g. At the completion of this test, the transmitter shall be subjected to the tests specified in 4.6.3 and 4.6.4. Scale and friction errors shall not exceed the tolerances specified in table I and II, respectively.

4.6.14.1 Vibration endurance. The transmitter shall be subjected to an endurance vibration cycling test for 40 hours in each of the three mutually perpendicular axes. The vibration spectrum shall be in accordance with MIL-STD-810C, method 514, equipment category (b), curve L, and the duration shall be in accordance with the figure titled "Logarithmic cycling rate:." During this test, the scale error test shall be conducted at 10, 20, 40, 60, and 100 percent of full scale. Scale error shall not exceed the tolerance specified in table II for vibration at 20g. After completion of this test, the scale error test
shall again be conducted. Errors shall not exceed the tolerances specified in table I for high and low temperatures. The transmitter shall be completely disassembled and examined for damage or wear that might result in failure during the design life of the instrument.

4.6.15 Pressure drop. The pressure drop across the transmitter shall be determined at the flow rates specified in table III. The pressure drop shall be recorded with the transmitter both energized and unenergized. The pressure drop shall be obtained by sensing the live pressure at the inlet and outlet end of the flowmeter and then obtaining the difference between the two readings. The pressure drop shall not exceed the values specified in table III.

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4.6.16 Low temperature exposure. The transmitter shall be placed in a test chamber and the chamber cooled to and maintained at a temperature of -62°C for a period of 72 hours at which time the equipment shall be inspected for deterioration. The temperature of the chamber shall then be raised to -54°C and maintained for an additional 24-hour period. At the end of the 24-hour period and while still at -54°C, the transmitter shall be subjected to the scale error and friction error tests. The transmitter shall then be maintained at room temperature for a period of not less than four hours after which it shall be subjected to the scale error and friction error tests at room temperature. The transmitter shall be examined and there shall be no deterioration of nor damage to the transmitter as a result of the extreme temperature exposure. The errors shall not exceed those specified in tables I and II.

4.6.17 Temperature-altitude. The transmitter shall be placed in a test chamber and the chamber cooled to and maintained at a temperature of -54°C ±2°C and at a pressure equivalent to 100,000 feet for a period of four hours. The transmitter shall be operated with no fuel flowing. Immediately after the four-hour period at the specified temperature and pressure, the transmitter shall be returned to normal room conditions and the scale error and friction error tests shall be conducted. The errors shall not exceed those specified in tables I and II for room-temperature conditions. This test may be combined with the test specified in 4.6.16. There shall be no damage to nor deterioration of the transmitter as a result of this test.

4.6.18 High temperature exposure. While inoperative, the transmitter shall be subjected to a temperature of 121°±2°C for a period of four hours. The temperature shall be raised to 190°±2°C for a period of one hour. This cycle shall be repeated for a total of 48 hours at 121°±2°C and 12 hours at 190°±2°C during which fluid may be inside the transmitter but not flowing. The transmitter shall be energized and operated for the last two temperature cycles after which it shall be operated at 140°±2°C for four hours at 1,000 pph with fuel flowing. The temperature of the fuel shall be 70°±2°C. The transmitter shall then be immediately tested for scale error and friction at 121°±2°C ambient air and the fuel at 50°±2°C. The tolerances shall not exceed those specified in tables I and II. The transmitter shall be examined and there shall be no evidence of failure.

4.6.19 Pressure leakage-high temperature. The transmitter shall be subjected to a pressure leakage at high temperature test as follows: A nonshock pressure shall be applied to the inlet connection with the outlet connection bled free of air and capped. The transmitter shall be heated in a 140°C chamber for 1/2 hour and a pressure of 1,000 psi shall then be applied rapidly five times. After the last application of pressure and while still at the test pressure and temperature, there shall be no indication of leakage for five minutes as checked by a pressure gage readable to 10 psi or by visual leakage of a colored test fluid.

4.6.20 Humidity. The transmitter shall be subjected to a humidity test in accordance with MIL-STD-810C, method 507, after which it shall be subjected to the tests specified in 4.6.3 and 4.6.4.

4.6.21 Fungus. The transmitter shall be subjected to a fungus test in accordance with MIL-STD-810C, method 508, procedure I. At the conclusion of this test, the transmitter shall be visually examined. There shall be no deterioration or corrosion of internal components as a result of this test.

4.6.22 Salt fog. The transmitter shall be subjected to a salt fog test in accordance with MIL-STD-810C, method 509, procedure I, for a period of 50 hours. The scale error and friction error tests shall be conducted before and after this test. The errors shall not exceed the low and high temperature tolerances. The transmitter shall also pass the test specified in 4.6.2.

4.6.23. Electromagnetic interference. The transmitter shall be subjected to electromagnetic conducted, radiated, and susceptibility tests in accordance with MIL-STD-462 for class ID equipment except the frequency range shall be from 0.15 to 150 mc.

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4.6.24 Shock. The shock test shall be conducted in accordance with MIL-STD-810C, method 516, procedure I.

4.6.25 Contaminated fuel operation. Test fuel containing dirt to the extent of 800 grams of foreign matter to 1,000 gallons of fuel shall be circulated through the transmitter at approximately 2,000 pph for 1 hour. Scale error and friction error tests shall then be conducted using the contaminated fuel. Operation shall be satisfactory and within the room temperature tolerances. The foreign matter shall consist of not less than 68 percent SiO2 and shall have a particle analysis as shown in table IV.

4.6.26 Frequency cycling. The master indicator need not be used for the cycling portion of this test. A second harmonic, or simulated receiver compatible with the transmitter will be adequate. A master power supply shall be used. The voltage shall be adjusted to 115±1V and the frequency cycled from 320 ±2 to 480 ±2 Hz in accordance with the following schedule with approximately half-scale flow. This constitutes one-cycle for a total time of 100 seconds.

a. Start cycle at 320 Hz, increase frequency to 480 Hz in 38 seconds linearly with time

b. Return frequency to 320 Hz in two seconds or less

c. Increase frequency to 48 Hz in 58 seconds linearly with time

d. Return frequency to 320 Hz in two seconds or less

The cycle shall be repeated continuously for ten hours at the rate of 36 cycles per hour. Power shall than be removed for a maximum of two hours and reapplied for one hour at 120 ±1V 320 ±2 Hz. Immediately following the one hour of operations at 120V 320 Hz, the transmitter shall be frequency cycled for one additional hour on the schedule indicated above, except that the frequency range shall be increased from 300 to 500 Hz. The transmitter shall then be subjected to the scale error test at room temperature. The scale error shall not exceed the tolerances specified in table I for room temperature.

4.6.27 Compatibility. The transmitter shall be subjected to the tests specified in 4.6.3, 4.6.9, and any other tests necessary to determine compliance with 3.4.1, 3.4.2, 3.4.3, 3.5.4, and 3.8.

4.6.28 Reliability

4.6.28.1 Qualification. The 11 transmitters specified in 4.4.1 shall be subjected to a reliability test in accordance with MIL-STD-781 for jet fighter aircraft, which shall demonstrate a minimum acceptable MTBF of 1,000 hours to a 90 percent confidence level.

4.6.28.1.2 Operation. During this test, fuel at room temperature shall be flowing through the transmitter at a rate from 5 to 90±5 percent of full scale and return at a rate of 2 ±1 cpm.

4.6.28.1.3 Temperature cycling. Unless otherwise specified or amended by the procuring activity due to the results of the thermal analysis, the temperature of the chamber in which the transmitters are tested shall be cycled as follows:

a. Starting at room temperature, decrease temperature to -54°C and stabilize for one hour with transmitters unenergized

b. Turn transmitters on, raise temperature to 125°C, and operate transmitters for three hours

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c. Turn transmitters off and lower chamber temperature to -54°C for one hour

d. Repeat steps b and c until reliability test requirement has been met. This schedule provides six 4-hour test cycles for each 24-hour period. In each 24-hour period the operational test time is 18 hours.

4.6.28.1.4 Vibration cycling. Throughout this test, the transmitter shall be vibrated for a period of ten minutes during each hour of "On" time at any nonresonant frequency between 20 and 60 Hz and at a peak acceleration of 2.2g 10 percent. During this test, the transmitter motor shall undergo voltage cycling from 65V 400 Hz to 126V 400 Hz linearly in time at a rate of 1 cycle every 4±1 minutes during operational "On" time starting with 115V 400 Hz.

4.6.28.1.5 Performance check. At each 200-hour interval and at the end of this test, the transmitters shall be subjected to the scale error, friction, and response time tests. During each scale error test, if any two test points exceed twice the allowable tolerances for high and low temperatures specified in tables I and II, one failure shall be charged against this test.

4.6.28.1.6 Failure criteria. For an accept decision, each transmitter shall be operated at least 600 hours. If a transmitter fails, entries shall be made on the appropriate data logs and the failed equipment shall be removed from the test and repaired with minimum interruption of the transmitters still under test. The procuring activity shall be immediately notified of the failure and the plan for correcting the deficiency. A detailed analysis of the failure will be submitted as soon as it has been completed. No more than a total of 12 failures will be permitted during this test. Acceptance shall be based on the accept-reject criteria specified in MIL-STD-781, test plan I.

4.6.29 Longevity. Testing on 2 of the 11 reliability test samples be continue until 2,500 hours have accumulated on each transmitter. The cause of any failures shall be determined. Any design flaws shall be corrected.

4.6.30 Maintainability. The contractor shall submit his maintainability program to the procuring activity for review and approval to determine that the requirements specified in 3.4.8 will be met.

4.7 Inspection of packaging. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C" as specified (see 6.2).

5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in Quantity Unit Pack (QUP) of one each. Each item will be provided a preservation method IC-1 In Accordance With (IAW) MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned IAW MIL-P-116, C-1 process

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved IAW MIL-P-116 and MIL-STD-2073-1 to provide a method IC-1.

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5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073-1.

5.2 Packaging. Packing shall be level "A," "B," or "C" as specified (see 6.2).

5.2.1 Container-cushioning. Unit container requirements shall conform to MIL-B-121, barrier material, unless otherwise specified by the contracting activity. Unit containers shall be large enough to allow for application of sufficient cushioning/wrap material shall be of sufficient density and thickness to protect the unit container from punctures. Intermediates/shipping containers shall be large enough to allow for application of sufficient cushioning material, between the unit container and intermediate/shipping container. Cushioning material shall be of sufficient density and thickness to ensure shock transmission does not exceed peak values in G's established for the item.

5.2.2 Level A. Fiberboard containers do not meet Level "A" container criteria.

5.2.3 Level B. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-1672, weather-resistant unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size and minimum tare and cube, consistent with the protection required. Closure shall be in accordance with appropriate PPP-B-1672 procedures, as specified by Special Packaging Instructions (SPI), and contracting activity.

5.2.4 Level C. Item will be packed in such a manner that will afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-2073-1.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked IAW special markings required by the contracting activity, SPI, and MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The TRU-6/A transmitter covered by this specification is intended for use in measuring and transmitting a signal proportional to the total mass rate of fuel flow to the main engine of jet aircraft.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. That the items specified in 4.5.3 be returned to the contractor for appropriate corrective action at the contractor's expense.

c. Levels of packaging and packing required and reinspection date marking (see section 5).

d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

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6.2.1 Reliability test program. Within 90 days after receipt of contract or purchase order and 30 days prior to the scheduled date of initiation of testing or at such later date as the contracting officer may authorize, the contractor will submit a detailed reliability test program to the procuring activity for approval. Testing will begin only after formal approval of the test program by the procuring activity.

6.3 First article. When the first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.2.

6.4 Subject term (keyword) listing.

Fuel
Synchro

6.5 Changes from previous issue. Marginal notations are not used in the revisions to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
AIR FORCE-99
NAVY-AS

Preparing Activity:
AIR FORCE-71

Project number:
6620-0537

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-26298F	94/09/30

3. DOCUMENT TITLE: Transmitter, Rate Of Flow TRU-6/A, 05,000-YPH

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
OC-ALC/TFCLA	405-736-5960	336-5960
c. ADDRESS (Include Zip Code) 3001 Staff Dr. TAFB OK 73145-5990	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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